**Example 3.66** If $A=\begin{pmatrix}1 & 8 & 3\\ 3 & 5 & 0\\ 8 & 7 & 6\end{pmatrix}$, $B=\begin{pmatrix}8 & -6 & -4\\ 2 & 11 & -3\\ 0 & 1 & 5\end{pmatrix}$, $C=\begin{pmatrix}5 & 3 & 0\\ -1 & -7 & 2\\ 1 & 4 & 3\end{pmatrix}$

compute the following : (i) $3A+2B-C$ (ii) $\frac{1}{2}A-\frac{3}{2}B$

***Solution*** (i) $3A+2B-C=3\begin{pmatrix}1 & 8 & 3\\ 3 & 5 & 0\\ 8 & 7 & 6\end{pmatrix}+2\begin{pmatrix}8 & -6 & -4\\ 2 & 11 & -3\\ 0 & 1 & 5\end{pmatrix}-\begin{pmatrix}5 & 3 & 0\\ -1 & -7 & 2\\ 1 & 4 & 3\end{pmatrix}$

$=\begin{pmatrix}3 & 24 & 9\\ 9 & 15 & 0\\ 24 & 21 & 18\end{pmatrix}+\begin{pmatrix}16 & -12 & -8\\ 4 & 22 & -6\\ 0 & 2 & 10\end{pmatrix}+\begin{pmatrix}-5 & -3 & 0\\ 1 & 7 & -2\\ -1 & -4 & -3\end{pmatrix}$

$=\begin{pmatrix}14 & 9 & 1\\ 14 & 44 & -8\\ 23 & 19 & 25\end{pmatrix}$

(ii) $\frac{1}{2}A-\frac{3}{2}B=\frac{1}{2}(A-3B)$

$=\frac{1}{2}\left(\begin{pmatrix}1 & 8 & 3\\ 3 & 5 & 0\\ 8 & 7 & 6\end{pmatrix}-3\begin{pmatrix}8 & -6 & -4\\ 2 & 11 & -3\\ 0 & 1 & 5\end{pmatrix}\right)$

$=\frac{1}{2}\left(\begin{pmatrix}1 & 8 & 3\\ 3 & 5 & 0\\ 8 & 7 & 6\end{pmatrix}+\begin{pmatrix}-24 & 18 & 12\\ -6 & -33 & 9\\ 0 & -3 & -15\end{pmatrix}\right)=\frac{1}{2}\begin{pmatrix}-23 & 26 & 15\\ -3 & -28 & 9\\ 8 & 4 & -9\end{pmatrix}$

$=\begin{pmatrix}-\frac{23}{2} & 13 & \frac{15}{2}\\ -\frac{3}{2} & -14 & \frac{9}{2}\\ 4 & 2 & -\frac{9}{2}\end{pmatrix}$

## Exercise 3.18

1. If $A=\begin{pmatrix}1 & 9\\ 3 & 4\\ 8 & -3\end{pmatrix}$, $B=\begin{pmatrix}5 & 7\\ 3 & 3\\ 1 & 0\end{pmatrix}$ then verify that

   (i) $A+B=B+A$ (ii) $A+(-A)=(-A)+A=O$.

2. If $A=\begin{pmatrix}4 & 3 & 1\\ 2 & 3 & -8\\ 1 & 0 & -4\end{pmatrix}$, $B=\begin{pmatrix}2 & 3 & 4\\ 1 & 9 & 2\\ -7 & 1 & -1\end{pmatrix}$ and $C=\begin{pmatrix}8 & 3 & 4\\ 1 & -2 & 3\\ 2 & 4 & -1\end{pmatrix}$ then verify that

   $A+(B+C)=(A+B)+C$.



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